Lösung: Vereinfachen komplexer Ausdrücke & Potenzen

a) $(1+i)^{804} = \ldots$ lässt sich in Exponentialdarst. leichter berechnen.

Betrag: $|1+i| = \sqrt{1^2+1^2} = \sqrt{2}$

Winkel: $\arg(z) = \arctan\left(\frac{1}{1}\right) = \arctan(1) = \pi/4 \quad (= 45°)$

(..., da im 1. Quadrant

$\Leftrightarrow \quad 1+i = \sqrt{2} \cdot e^{i\pi/4}$

$$(1+i)^{804} = \left(\sqrt{2} \cdot e^{i\pi/4}\right)^{804} = \left(\sqrt{2}\right)^{804} \cdot \left(e^{i\pi/4}\right)^{804}$$

$$= 2^{1/2 \cdot 804} \cdot e^{i 804 \cdot \pi/4} = 2^{402} \cdot e^{i 201 \cdot \pi}$$

$$= 2^{402} \cdot e^{i(200+1)\pi} = 2^{401} \cdot \underbrace{e^{i200\pi}}_{= e^{i2\pi \cdot 100}} \cdot e^{i\pi}$$

$$= 2^{402} \cdot \underbrace{\left(e^{i2\pi}\right)^{100}}_{=1} \cdot e^{i\pi}$$

$$= 2^{402} \; 1^{200} \cdot \underbrace{e^{i\pi}}_{=-1} = \underline{\underline{-2^{402}}}$$

b) $e^{in\pi} = \left(e^{i\pi}\right)^n = (-1)^n$

c) $\left(\sqrt[5]{3} \cdot \left(\cos\left(\frac{\pi}{20}\right) + i \sin\left(\frac{\pi}{20}\right)\right)\right)^5 = \left(\sqrt[5]{3} \cdot e^{i\pi/20}\right)^5$

$$= \left(\sqrt[5]{3}\right)^5 \cdot \left(e^{i\pi/20}\right)^5 = 3 \; e^{i5 \cdot \pi/20} = 3 \; e^{i\pi/4}$$

$$= 3 \left(\cos(\pi/4) + i \sin(\pi/4)\right) = 3 \left(\frac{1}{\sqrt{2}} + i \frac{1}{\sqrt{2}}\right)$$

$$= \underline{\underline{\frac{3}{\sqrt{2}} + i \frac{3}{\sqrt{2}}}}$$

d) $e^{-i\,3/4\pi} + 2 \cdot e^{i\,3/2\pi} = \ldots$ ← Addition ist nur in kartesischer Darstellung möglich.

1. Term: $e^{-i\,3/4\pi} = \cos(-3/4\pi) + i \sin(-3/4\pi)$

$= \cos(3/4\pi) - i \sin(3/4\pi)$

$= -1/\sqrt{2} - i\,(+1/\sqrt{2}) = -\frac{1}{\sqrt{2}} + i\,\frac{1}{\sqrt{2}}$

2. Term: $2 \cdot e^{i\,3/2\pi} = 2\,(\cos(\frac{3}{2}\pi) + i \sin(3/2\pi))$

$= 2\,(0 + i\,(-1)) = -2i$

$\Rightarrow \quad e^{-i\,3/4\pi} + 2 \cdot e^{i\,3/2\pi} = -\frac{1}{\sqrt{2}} + i\,\frac{1}{\sqrt{2}} - 2i = -\frac{1}{\sqrt{2}} - i\,(2 - \frac{1}{\sqrt{2}})$

$= \underline{\underline{-\frac{1}{\sqrt{2}} + i\,\frac{1 - \sqrt{2} \cdot 2}{\sqrt{2}}}}$